

NULL FUNCTIONS • Exercise 5

Table 1: Employees

Q1. Show all employees with their salary. If salary is NULL, display 0

Expected Columns: employee_id, name, salary - with default

```
SELECT employee_id,  
        name,
```

```
IF NULL (salary) AS salary_with_default  
FROM employees;
```

```
SELECT employee_id,  
        name
```

```
IF NULL (salary, 0) AS salary_with_default  
FROM employees;
```

Outcome table:

employee_id	name	salary_with_default
2	Bob	8
3 4	Donald Donald	0

Q2: Show employees names with their department. If department is NULL, show "Not Assigned"

```
SELECT employee_id,  
        name,  
department_name
```

```
IF NULL (department, "Not assigned") AS department_name
```

Outcome table

employee_id	name	department_name
3	Charlie	Not assigned

Table 2: Orders

Q3. Find orders with NULL customer_id using ISNULL().

```
SELECT order_id,  
       customer_id  
FROM orders  
WHERE customer_id IS NULL;
```

Outcome Results:

order_id	customer_id
103	NULL

Q4. Show all orders. If delivery_date is NULL, show 'Pending'

```
SELECT order_id,  
       customer_id  
FROM orders  
WHERE
```

```
SELECT order_id,  
       customer_id,  
       COALESCE(delivery_date, Pending)  
       AS delivery_status.  
FROM orders;
```

Outcome table:

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Table 3: Students

Q5. Show all students and their grades. Replace NULL with 0
SELECT student_id,
name
IFNULL(grade, 0) AS final_grade
FROM students;

Outcome table:

student_id	name	grade
2	Maya	0

Q6. Count students who haven't been graded.
SELECT COUNT(*) AS not_graded_count
FROM students
WHERE grade is NULL;

Outcome table

Not_graded_count
1

Table 4: Products

Q7. Show product name, price and final price after discount (assume 0 if discount is NULL).
SELECT product_id,
name
IFNULL(discount, 0) AS final_price
FROM products;

Outcome table

product_id	name	final-price
501	Keyboard	0
502	Mouse	5
503	Monitor	0

Table 5: Customers

Q8: Count how many customers have no email.

SELECT COUNT(*) AS missing_email_count.

Outcome table

missing_email_count
2

Q9: Show all customers with email. If NULL, display "No email"

SELECT customer_id,
name

FROM customers

WHERE email IS NULL;

Outcome Results

customer_id	name	email-display
1	Linda	NULL
3	Nia	NULL

Table 6: Payments

Q10: Show payments details with method replaced by "Unknown" if NULL

SELECT payment_id,

COALESCE(method, 'Unknown') AS method_display,
status

FROM payments

Outcome table

payment_id	method_display	status
301	credit	Unknown
302	Paypal	Success
303	Unknown	Failed

Table 7: Inventory

Q1 Show items and their quantity (0 if NULL)

```
SELECT item_id,  
       item_name,  
       COALESCE(quantity, 0) AS quantity_checked  
FROM inventory;
```

Outcome table.

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8 : Employee_Extra

Q12. Show employee ID and the first available value among bonus or commission.

```
SELECT emp_id,  
       COALESCE(bonus, commission) AS first_available_reward  
FROM Employee_Extra;
```

Table outcome:

Emp_id	First_available_reward
1	300
2	100
3	NULL

Table 9 : Classes

Q13. Count Classes that don't have a room assigned.

```
SELECT COUNT(*) AS no_room_count  
FROM classes  
WHERE room IS NULL
```

Table outcome:

no_room_count
2

Q14. Show attendance records with status. Replace NULL with "Not Marked"

```
SELECT student_id,  
       date,  
       COALESCE(status, "Not Marked") AS attendance_status  
FROM Attendance;
```

Outcome Results.

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent.

Table 11: Bank Accounts
Q15. Show account ID, account_type (or Unknown) and balance (or 0)

```
SELECT account_id,  
       COALESCE(account_type, Unknown) AS type_display,  
       COALESCE(balance, 0) AS balance_checked.  
FROM Bank_accounts;
```

Outcome table.

account_id	type_display	balance_checked.
A1	savings.	0
A2	current	5000
A3	Unknown	2000

Table 12: Projects.

SELECT project_id,
title,
COALESCE(start date, TBD) AS start display..

Outcome Results

project_id	title	start display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table B: Reviews

Display reviews showing comment (or No comment) and rating or (0).

SELECT review_id,
product_id
COALESCE(comment, 'No comment') AS comment display
COALESCE(rating, 0) AS rating-display.

FROM Reviews.;

Outcome table

review_id	product_id	comment-display	rating-display
1	501	great product	4
2	502	No comment	0
3	503	Works fine.	3

Table 14 : Suppliers

Q18 Show the supplier contact number. Use COALESCE(phone, alt_phone, No contact)

```
SELECT supplier_id,  
       name,
```

```
       COALESCE(phone, alt_phone, No contact) AS contact_number
```

Outcome table

supplier_id	name	contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	Value Source	NO Contact

Table 15 : User settings

```
SELECT user_id,
```

```
       COALESCE(theme, 'Light') AS theme_set,
```

```
       COALESCE(language, 'English') AS language,
```

```
       COALESCE(timezone, 'UTC') AS timezone_set
```

```
FROM user_settings
```

Outcome Results

user_id	theme_set	language_set	timezone_set
1	light	English	UTC
2	dark	English	UTC+1
3	light	English	UTC

Table 16: Maintenance

Q20: Show maintenance log with

SELECT record_id,
machine_id,
COALESCE (issue, 'Unknown Issue') AS issue_log.
COALESCE (technician, 'Not assigned') AS technician_name.

Outcome Results

record_id	machine_id	issue_log	technician
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed.	Alex